

```

127.         # Check each box:
128.         for boxx in (0, 3, 6):
129.             for boxy in (0, 3, 6):
130.                 boxNumbers = []
131.                 for x in range(BOX_LENGTH):
132.                     for y in range(BOX_LENGTH):
133.                         number = self.grid[(boxx + x, boxy + y)]
134.                         boxNumbers.append(number)
135.                 if not self._isCompleteSetOfNumbers(boxNumbers):
136.                     return False
137.
138.         return True
139.
140.
141. print('''Sudoku Puzzle, by Al Sweigart al@inventwithpython.com
142.
143. Sudoku is a number placement logic puzzle game. A Sudoku grid is a 9x9
144. grid of numbers. Try to place numbers in the grid such that every row,
145. column, and 3x3 box has the numbers 1 through 9 once and only once.
146.
147. For example, here is a starting Sudoku grid and its solved form:
148.
149.      5 3 . | . 7 . | . . .      5 3 4 | 6 7 8 | 9 1 2
150.      6 . . | 1 9 5 | . . .      6 7 2 | 1 9 5 | 3 4 8
151.      . 9 8 | . . . | . 6 .      1 9 8 | 3 4 2 | 5 6 7
152.      -----+-----+-----      -----+-----+-----
153.      8 . . | . 6 . | . . 3      8 5 9 | 7 6 1 | 4 2 3
154.      4 . . | 8 . 3 | . . 1 --> 4 2 6 | 8 5 3 | 7 9 1
155.      7 . . | . 2 . | . . 6      7 1 3 | 9 2 4 | 8 5 6
156.      -----+-----+-----      -----+-----+-----
157.      . 6 . | . . . | 2 8 .      9 6 1 | 5 3 7 | 2 8 4
158.      . . . | 4 1 9 | . . 5      2 8 7 | 4 1 9 | 6 3 5
159.      . . . | . 8 . | . 7 9      3 4 5 | 2 8 6 | 1 7 9
160. ''')
161. input('Press Enter to begin...')
162.
163.
164. # Load the sudokupuzzles.txt file:
165. with open('sudokupuzzles.txt') as puzzleFile:
166.     puzzles = puzzleFile.readlines()
167.
168. # Remove the newlines at the end of each puzzle:
169. for i, puzzle in enumerate(puzzles):
170.     puzzles[i] = puzzle.strip()
171.
172. grid = SudokuGrid(random.choice(puzzles))
173.
174. while True: # Main game loop.
175.     grid.display()
176.
177.     # Check if the puzzle is solved.
178.     if grid.isSolved():
179.         print('Congratulations! You solved the puzzle!')
180.         print('Thanks for playing!')
181.         sys.exit()

```